生醫量測系統設計

Design of Biomedical Measurement Systems

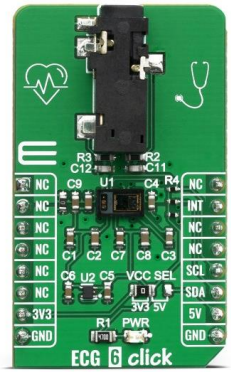
LAB 1

ECG&PPG signal extract

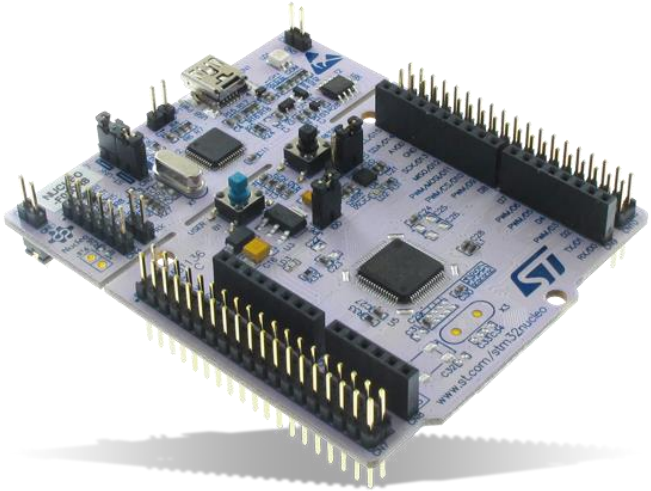
Hardware

- The hardware used in this project is as below.

1. ECG 6 click

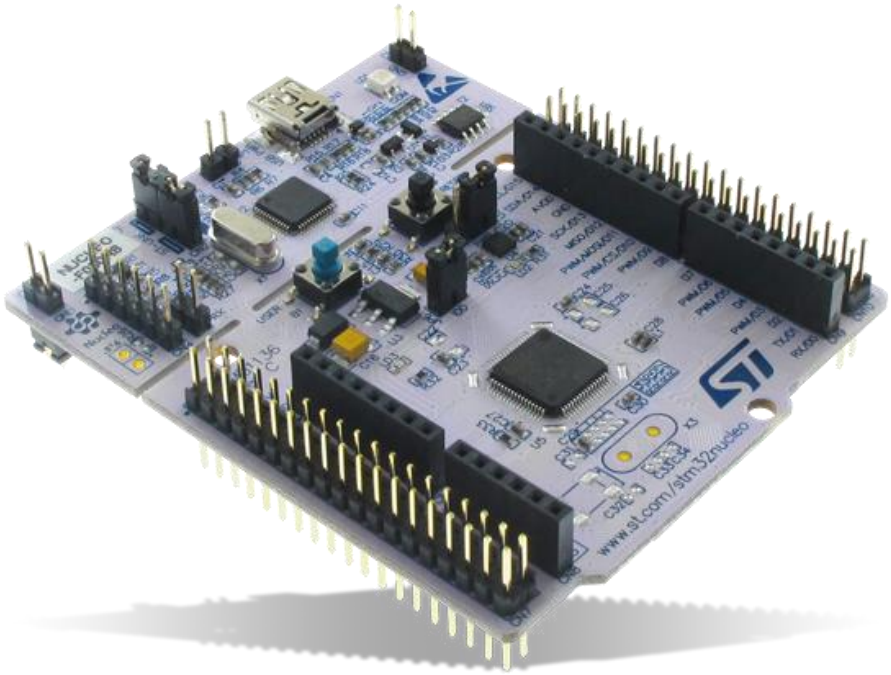


2. NUCLEO-F401RE



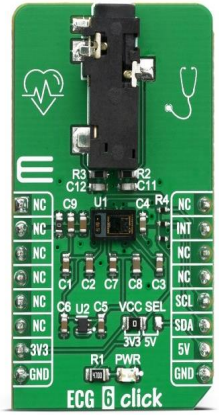
NUCLEO-F401RE

- Use STM32F401RE chip.
- Chip programmer is already on this dev-kit.

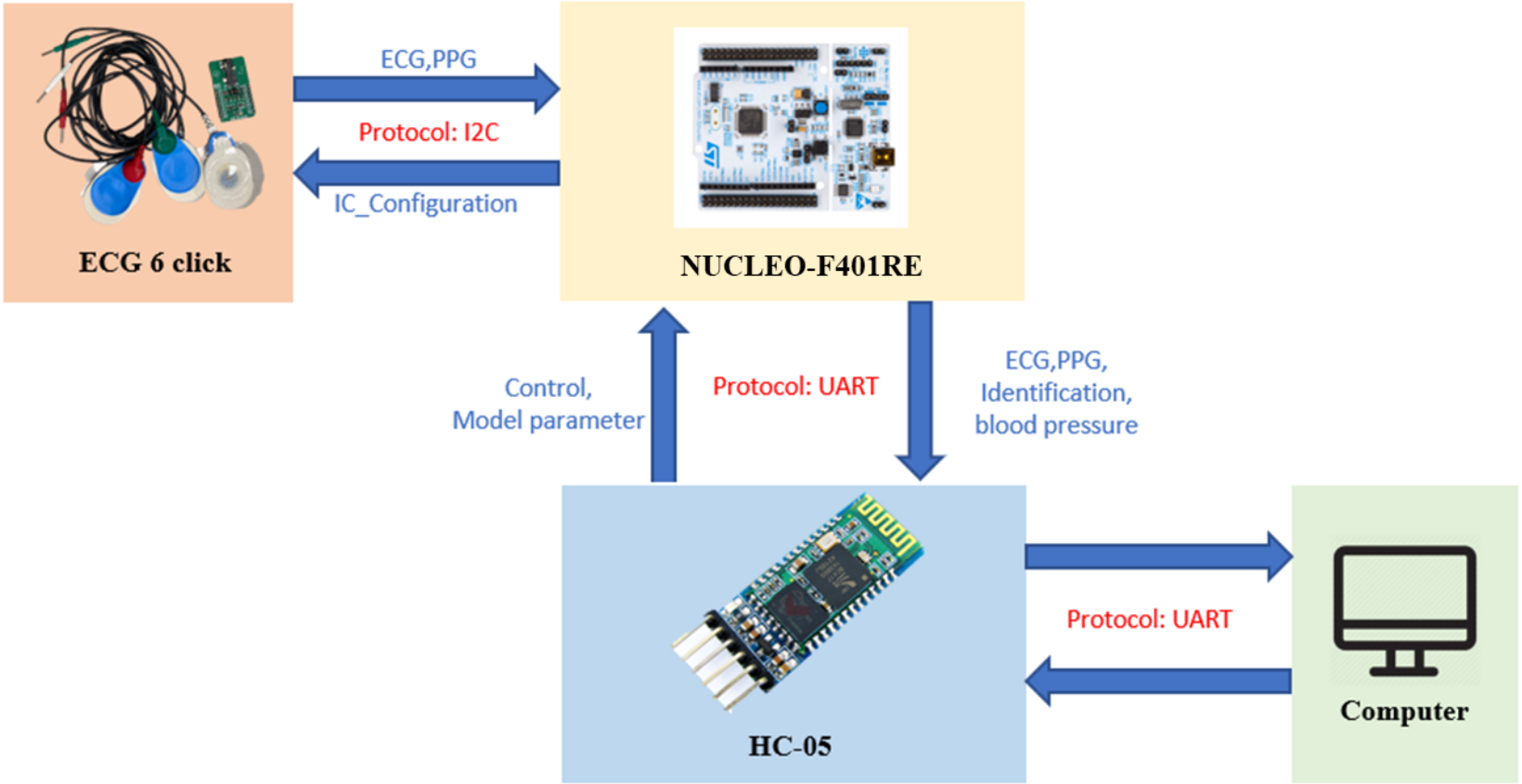


ECG 6 click

- ECG 6 click is a module to collect ECG and PPG data.
- The module includes MAX86150, LEDs, and a 3.5mm connector.



System Architecture



Development Environment

- STM32CubeIDE
- ANACONDA
- VsCode

Development Environment

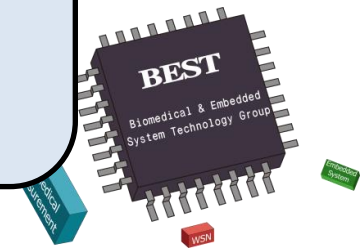
STM32 CubeIDE

Code Generation and Compile and Burn

ANACONDA.

Build Virtual Environment

Code Editor



ANACONDA

- Build Environment
  - conda create --name python38 python=3.8

```
(base) C:\Users\sampaulyi>conda create --name python38 python=3.8
```

  - Enter y to proceed

```
The following packages will be downloaded:

    package                    |            build
    ---------------------------|-----------------
    ca-certificates-2021.9.30  |       haa95532_1         116 KB
    certifi-2021.10.8          |   py38haa95532_0         152 KB
    openssl-1.1.11             |       h2bbff1b_0         4.8 MB
    python-3.8.12              |       h6244533_0        16.0 MB
    setuptools-58.0.4          |   py38haa95532_0         779 KB
    wheel-0.37.0               |     pyhd3eb1b0_1          33 KB
    wincertstore-0.2           |   py38haa95532_2          15 KB
    ------------------------------------------------------------
                                           Total:        21.9 MB

The following NEW packages will be INSTALLED:

  ca-certificates    pkgs/main/win-64::ca-certificates-2021.9.30-haa95532_1
  certifi            pkgs/main/win-64::certifi-2021.10.8-py38haa95532_0
  openssl            pkgs/main/win-64::openssl-1.1.11-h2bbff1b_0
  pip                pkgs/main/win-64::pip-21.0.1-py38haa95532_0
  python             pkgs/main/win-64::python-3.8.12-h6244533_0
  setuptools         pkgs/main/win-64::setuptools-58.0.4-py38haa95532_0
  sqlite             pkgs/main/win-64::sqlite-3.36.0-h2bbff1b_0
  vc                 pkgs/main/win-64::vc-14.2-h21ff451_1
  vs2015_runtime     pkgs/main/win-64::vs2015_runtime-14.27.29016-h5e58377_2
  wheel              pkgs/main/noarch::wheel-0.37.0-pyhd3eb1b0_1
  wincertstore       pkgs/main/win-64::wincertstore-0.2-py38haa95532_2


Proceed ([y]/n)?
```

ANACONDA

- Build Environment
  - conda info --env (you will see the python38)

```
(base) C:\Users\sampaulyi>conda info --env
# conda environments:
#
base                  *  C:\Users\sampaulyi\anaconda3
py38                     C:\Users\sampaulyi\anaconda3\envs\py38
python38                 C:\Users\sampaulyi\anaconda3\envs\python38
```

  - conda activate python38 (base will change to python38)

```
(base) C:\Users\sampaulyi>conda activate python38
```

```
(python38) C:\Users\sampaulyi>
```

ANACONDA

- After entering the virtual environment, enter the following command to complete the module installation.
  - pip install scikit-learn
  - **pip install pyserial**
  - pip install numpy
  - pip install PyQt5
  - pip install matpltlib
- After entering the command, you will receive a message as below.

```
Successfully installed joblib-1.1.0 numpy-1.21.2 scikit-learn-1.0 scipy-1.7.1 threadpoolctl-3.0.0
```

ANACONDA

- Use “cd” command to enter the directory.

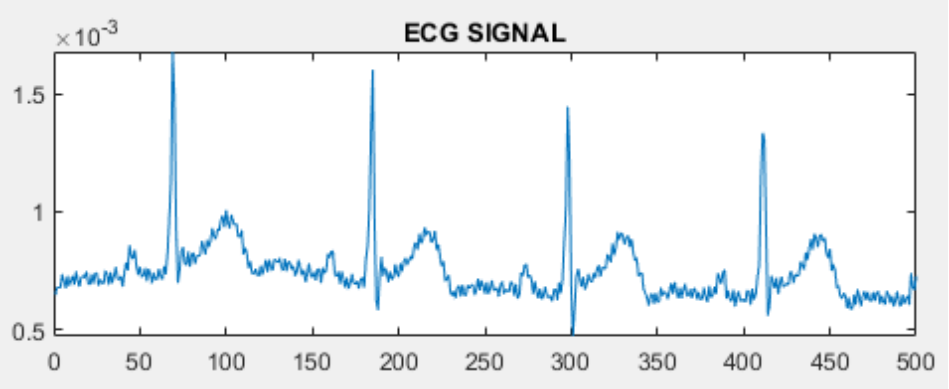
```
C:\>cd \Users\leo\Desktop\ecg\ecg_project_tech
```

- Use “python” command to execute the python code.

```
python lab_1.py
```

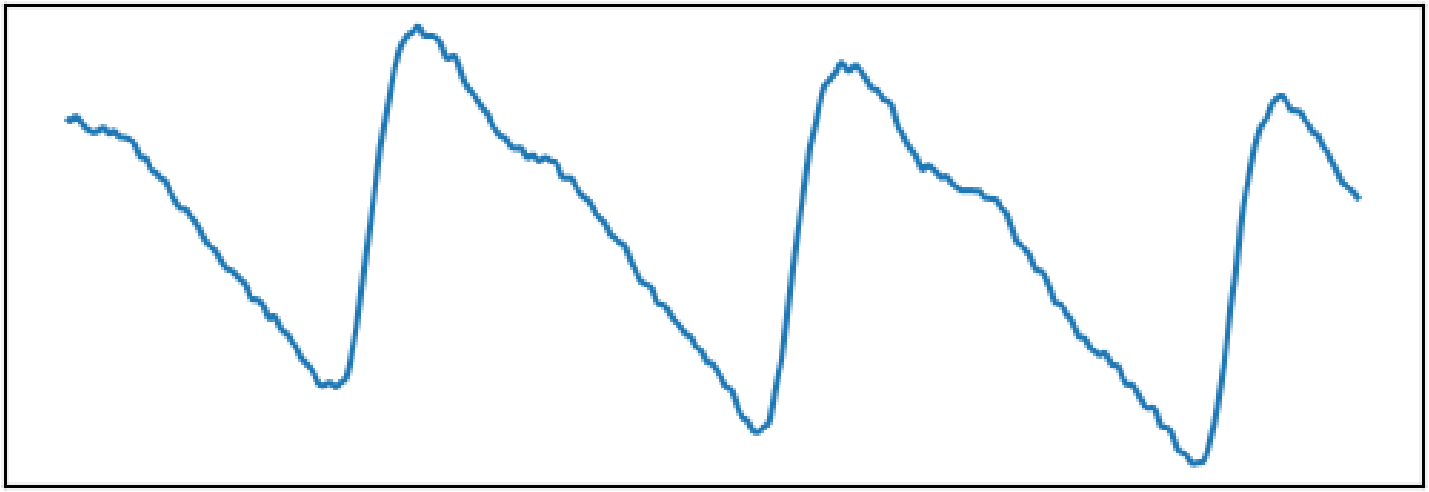
Electrocardiography

- Electrocardiography(ECG) is a graph of heart electrical signals. In this project, we will use lead 1.
- You can get lead 1 ECG by placing electrodes on the left arm, right arm, and left foot.



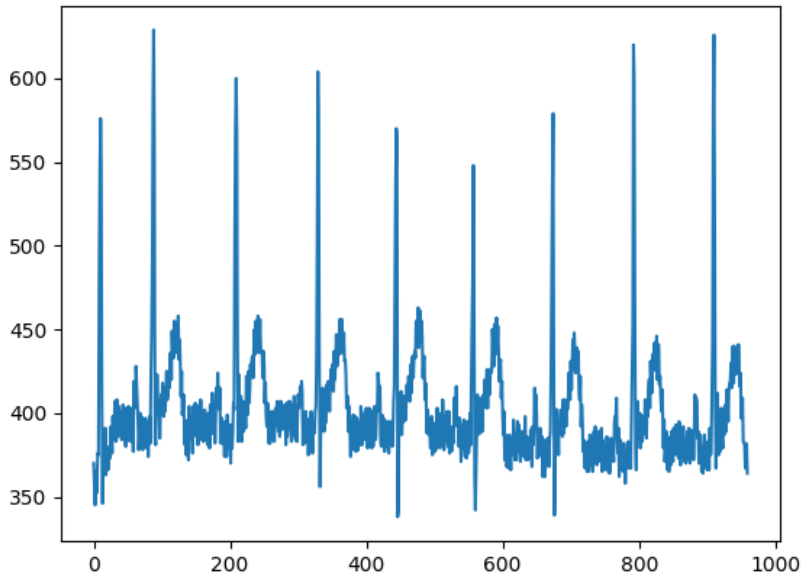
Photoplethysmography

- Photoplethysmography(PPG) is a method to measure the blood volume in blood vessels. We can use LED and photodiode to obtain the signal.



Data Preprocessing

- After receiving the data from MAX86150, the data needs to be processed first. This step calls “preprocessing”.

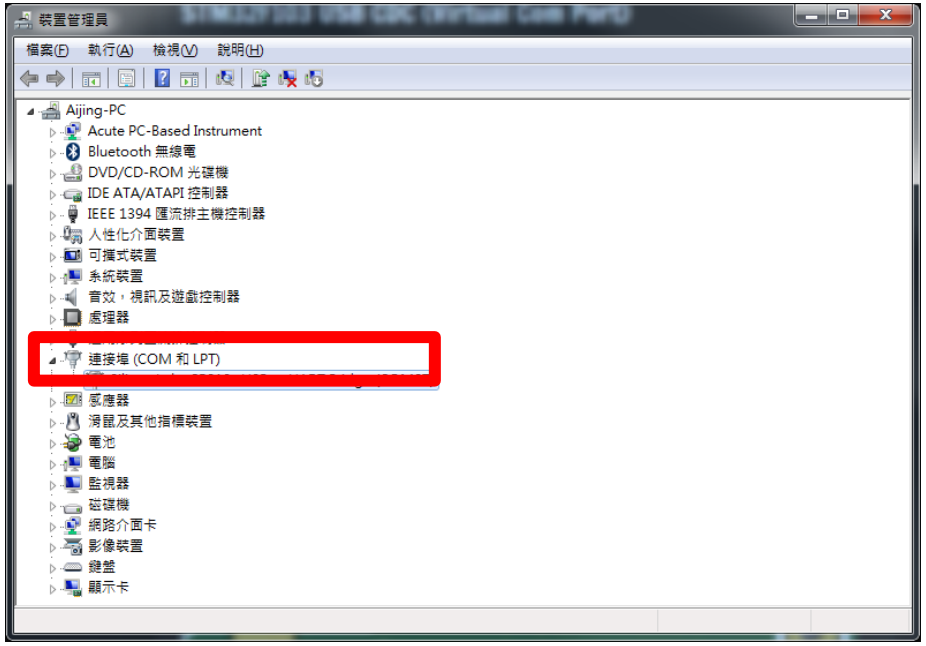


Filtering

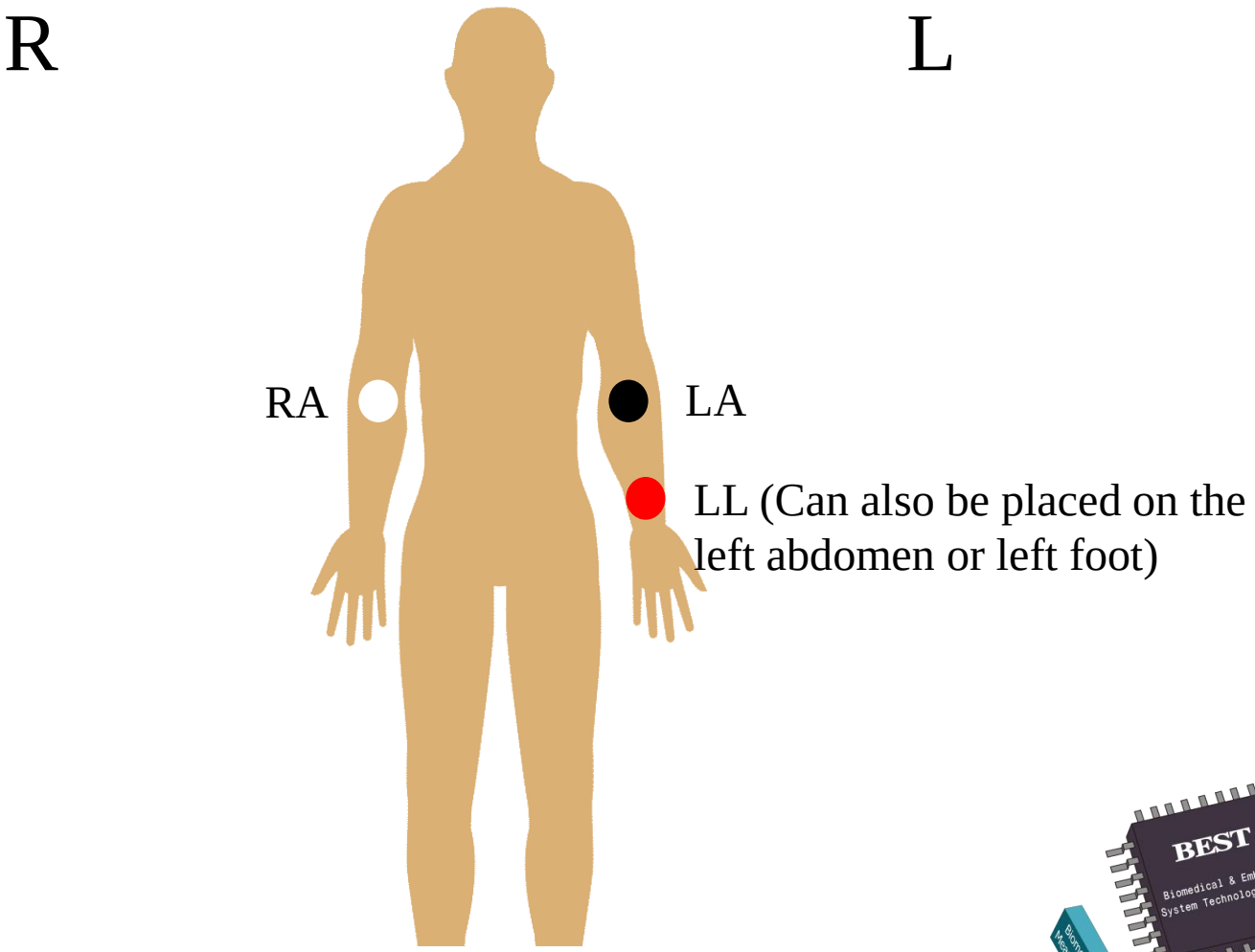
- Because ECG will be s with by environmental signals (e.g. muscle signals), filtering is an important part of preprocessing.

COM port

- Directly connect using USB
  - From 'Control Panel' to 'Device Manager'
  - Locate 'Ports (COM & LPT)' and check



ECG electrode position



Assignment

1. Use matplotlib.pyplot to plot the data on the screen.
2. Save ECG and PPG data in .txt.
3. Calculate Pulse Rate.